

Method of Data Summarization

$$(G.M)^2 = A.M \times H.M$$

* Measure of central tendency

1) A.M Individual series: $(\bar{X}) = \frac{\sum X}{n}$

Discrete series: $(\bar{X}) = \frac{\sum fX}{N}$

Continuous series: $(\bar{X}) = \frac{\sum fX}{N}$

where, X = mid points of classes:

2) G.M Individual series: $G.M = \text{Antilog} \left[\frac{\sum \log X}{n} \right]$

Discrete series: $G.M = \text{Antilog} \left[\frac{\sum f \log X}{n} \right]$

Continuous series: $G.M = \text{Antilog} \left[\frac{\sum f \log X}{n} \right]$

where, X = mid point of classes

3) H.M Individual series: $H.M = \frac{n}{\sum \frac{1}{X}}$

Discrete series: $H.M = \frac{N}{\sum f \cdot \frac{1}{X}}$

Continuous series: $H.M = \frac{N}{\sum f \cdot \frac{1}{X}}$

4) Median Individual series:

$$Md = \text{value of } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ item}$$

Discrete series:

$$Md = \text{value of } \left(\frac{N+1}{2}\right)^{\text{th}} \text{ item}$$

Continuous series:-

$$Md = l + \frac{\frac{N}{2} - cf}{f} \times h$$

$$Q_1 = \text{value of } \left(\frac{n+1}{4}\right)^{\text{th}} \text{ item}$$

$$Q_2 = \text{value of } 2\left(\frac{n+1}{4}\right)^{\text{th}} \text{ item}$$

$$Q_3 = \text{value of } 3\left(\frac{n+1}{4}\right)^{\text{th}} \text{ item}$$

5) Mode: Individual series:
(uni-modal) M_o = most repeated

Discrete series:

M_o = which have highest frequency

Continuous series:

$$M_o = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

(Bi-modal) $M_o = 3Md - 2\bar{X}$

a) Mean Deviation:

• Individual series:

$$\text{M.D from mean} = \frac{\sum |X - \bar{X}|}{n}$$

$$\text{M.D from median} = \frac{\sum |X - Md|}{n}$$

$$\text{M.D from mode} = \frac{\sum |X - Mo|}{n}$$

• Discrete series:

$$\text{M.D from mean} = \frac{\sum f \cdot |X - \bar{X}|}{N}$$

$$\text{M.D from median} = \frac{\sum f \cdot |X - Md|}{N}$$

$$\text{M.D from mode} = \frac{\sum f \cdot |X - Mo|}{N}$$

• Continuous series:

$$\text{M.D from mean} = \frac{\sum f \cdot |X - \bar{X}|}{N}$$

$$\text{M.D from median} = \frac{\sum f \cdot |X - Md|}{N}$$

$$\text{M.D from mode} = \frac{\sum f \cdot |X - Mo|}{N}$$

$$\text{Coefficient of M.D from mean} = \frac{\text{M.D from mean}}{\text{mean}}$$

$$\text{Coeff. of M.D from median} = \frac{\text{M.D from median}}{\text{median}}$$

$$\text{Coeffi. of M.D from mode} = \frac{\text{M.D from mode}}{\text{mode}}$$

7) Standard Deviation:

- Direct method:

$$\sigma = \sqrt{\frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2}$$

- Discrete series:

$$\sigma = \sqrt{\frac{\sum fX^2}{N} - \left(\frac{\sum fX}{N}\right)^2}$$

- Continuous series:

$$\sigma = \sqrt{\frac{\sum fX^2}{N} - \left(\frac{\sum fX}{N}\right)^2}$$

∴ Variance = σ^2

• Coefficient of variation = $\frac{\sigma}{\bar{X}} \times 100$

8) Skewness (Test of symmetry)

- Karl Pearson's coefficient of skewness:

$$S_k = \frac{\bar{X} - M_0}{\sigma}$$

$$S_k = \frac{3(\bar{X} - M_2)}{\sigma}$$

(Value of $S_k = -3$ to $+3$)

- Bowley's coefficient of skewness:

$$S_k = \frac{Q_3 + Q_1 - 2M_d}{Q_3 - Q_1}$$

(Value of $S_k = -1$ to $+1$)

- * $S_k = 0$ (symmetrical)
- * $S_k = +ve$ (+vely skewed)
- * $S_k = -ve$ (-vely skewed)

- Kurtosis (Normality test)

$$K = \frac{Q_3 - Q_1}{2(P_{90} - P_{10})}$$

$$K = 0.263 \text{ (mesokurtic)}$$

$$K > 0.263 \text{ (leptokurtic)}$$

$$K < 0.263 \text{ (platykurtic)}$$

$$P_{90} = \frac{90N}{100}$$

$$P_{10} = \frac{10N}{100}$$

Bmp Correlation

- 1) Karl Pearson's correlation coefficient:

Direct method:

$$r = \frac{n \sum XY - \sum X \cdot \sum Y}{\sqrt{n \sum X^2 - (\sum X)^2} \sqrt{n \sum Y^2 - (\sum Y)^2}}$$

Short cut method:

$$r = \frac{n \sum UV - \sum U \sum V}{\sqrt{n \sum U^2 - (\sum U)^2} \sqrt{n \sum V^2 - (\sum V)^2}}$$

gurukul

$$U = X - a$$

$$V = Y - b$$

Conclusion

$r = 0.001$ to $0.499 \rightarrow +ve$ correlation

$r = 0.5$ to $0.699 \rightarrow$ low degree of +ve correlation

$r = 0.7$ to $0.999 \rightarrow$ high degree of " "

$r = +1 \rightarrow$ perfect +ve correlation

$r = -0.001$ to $-0.499 \rightarrow -ve$ correlation

$r = -0.5$ to $-0.699 \rightarrow$ low " "

$r = -0.7$ to $-0.999 \rightarrow$ high degree -ve correlation

$r = -1 \rightarrow$ perfect -ve correlation

* Probable Error (P.E)

$$P.E = 0.6745 \times \frac{1-r^2}{\sqrt{n}}$$

r = correlation coefficient

n = number of pair obs.

Conclusion

$r > 6P.E \rightarrow r$ is significant (effective)

$r \leq 6P.E \rightarrow r$ is not " (not ")

2) Spearman's Rank Correlation Coefficient

Case I: (values not repeated)

Give rank '1' to highest value and so on,

$$r = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

where, $d = R_1 - R_2$

R_1 = Rank of values of 1st variable

R_2 = Rank of values of 2nd variable.
 n = no. of pair obs.

Case II: values repeated

If any value in the same variable repeats two or more than two times then give average equal rank to them acc. to positions which they take.

Then,

$$r = 1 - \frac{6 \left[\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \dots \right]}{n(n^2-1)}$$

where, $d = R_1 - R_2$

n = number of pair obs.

m_1, m_2 = repetition number of repeated value

3) Kendell Tau Correlation

Steps:

- 1) Rank the values of two variables X and Y .
- 2) Arrange the rank of 1st variable X in natural sequence and rank of 2nd variable Y in correspondingly.
- 3) Find the sum of all scores ' S '.
- 4) Then Kendell Tau Correlation coefficient is,

$$\tau = \frac{2S}{n(n-1)}$$

τ = Kendell Tau rank correlation coefficient

S = Sum of all scores

n = number of pair obs.